OBJECT ORIENTED PROGRAMMING
LAB ASSIGNMENT-6

(Topics – Introduction to Polymorphism, Function overloading)

Programming Questions:

Q1. Create a class Print with overloaded **display()** functions to display an integer, a double value, and a string.

Q2. Create a class Area with overloaded **calculate()** functions to calculate the area of a square, rectangle, and circle.

Q3. **Calculate Sum**
Create a class Calculator with overloaded functions named **sum():**

- sum(int a, int b) — calculate the sum of two integers.
- sum(int a, int b, int c) — calculate the sum of three integers.
- sum(double a, double b) — calculate the sum of two decimal numbers.

Demonstrate all three overloaded functions using an object of the class.

Q4. Create a class with overloaded **result()** functions to display marks in one subject, calculate the average of two subjects, and calculate the average of three subjects.

Q5. **Find the Maximum (Moderate)**
Create a class Maximum with overloaded functions named findMax():

- findMax(int a, int b) — find the greater of two integers.
- findMax(int a, int b, int c) — find the greatest of three integers.
- findMax(double a, double b) — find the greater of two decimal numbers.

Q6. **Shopping Bill Generator**
Create a class Bill with overloaded functions named calculateBill() to calculate:

- The bill for one item using quantity and unit price.
- The bill for two items using their quantities and prices.
- The bill for a bill amount after applying a discount percentage.

Use suitable parameter lists and return the final bill amount.